

Refrigeration and Air-Conditioning

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Air-Conditioning Systems

Human Comfort Conditions

- Depends upon physiological and psychological condition
- ASHRAE:- Americal Society of Heating, Refrigeration and air-conditioning Engineers
- As per ASHRAE:- “Human comfort is that conditions of mind, which expressed satisfaction with the thermal environment”

Skin temperature, $t_{\text{skin}} \approx 33.7^{\circ}\text{C}$
Core temperature, $t_{\text{core}} \approx 36.8^{\circ}\text{C}$

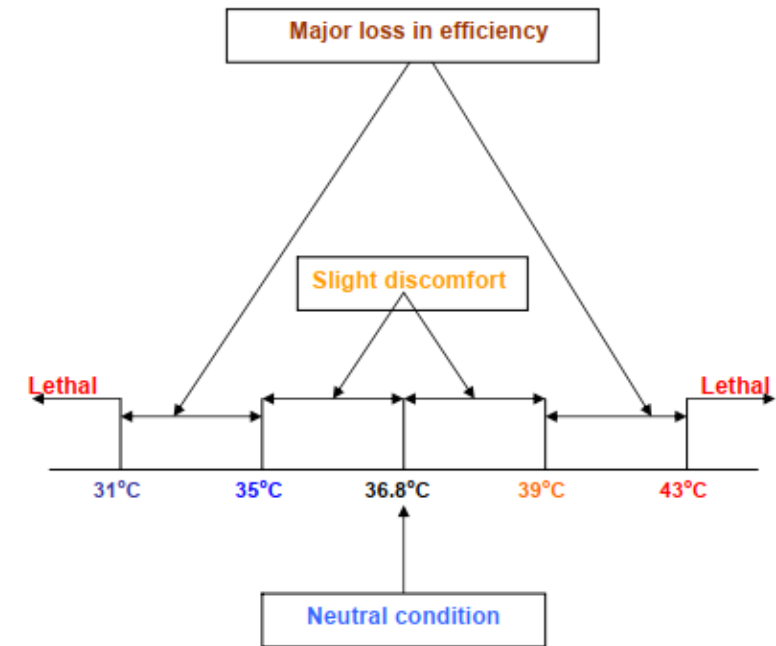


Fig.29.2: Affect of the variation of core temperature on a human being

Factors affecting Human Comfort

- **Effective Temperature:-** Function of DBT, RH and air velocity
- **Heat Production and regulation in human body:-** Sleeping – 60W/m^2 , Extensive work- 600W/m^2
- **Heat and moisture losses from human body**
- **Moisture content of air**
- **Quality and quantity of air:-** CO_2 content, Dust, odour
- **Air motion:-** Distribution of air, air velocity in occupied zone should not exceed 8m/min
- **Hot and cold surfaces:-** cold or hot objects in a conditioned space may cause discomfort to the occupants
- **Air stratification:-** Temperature gradient from floor to ceiling, air stratification should be minimum in occupied zone

Air-Conditioning Systems:- Classification

- 1. According to purpose**
 - a. Comfort A/C
 - b. Industrial A/C
- 2. According to season of year**
 - a. Winter A/C
 - b. Summer A/C
 - c. Year around A/C
- 3. According to arrangement of equipment**
 - a. Unitary A/C
 - b. Central A/C

According to purpose

1. Comfort A/C:- Air is brought to required DBT and RH for human health, comfort and efficiency

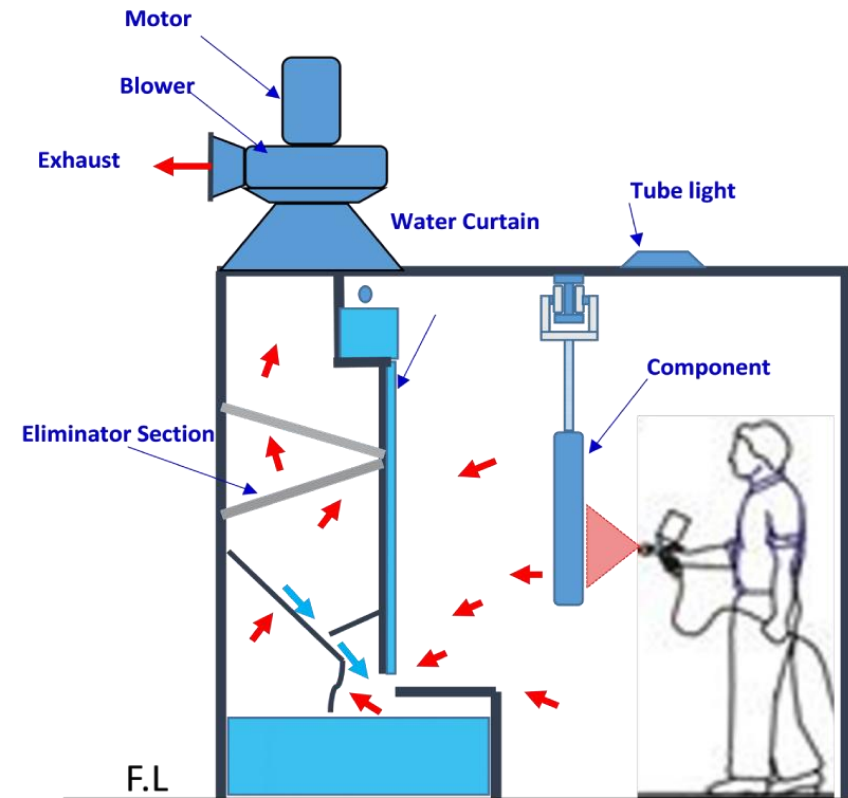
Ex:- Home, office, hospitals, shopping malls etc.

2. Industrial A/C:- DBT and RH of air kept constant for working of machine and manufacturing process

Ex: Textile mills, manufacturing plants, etc.

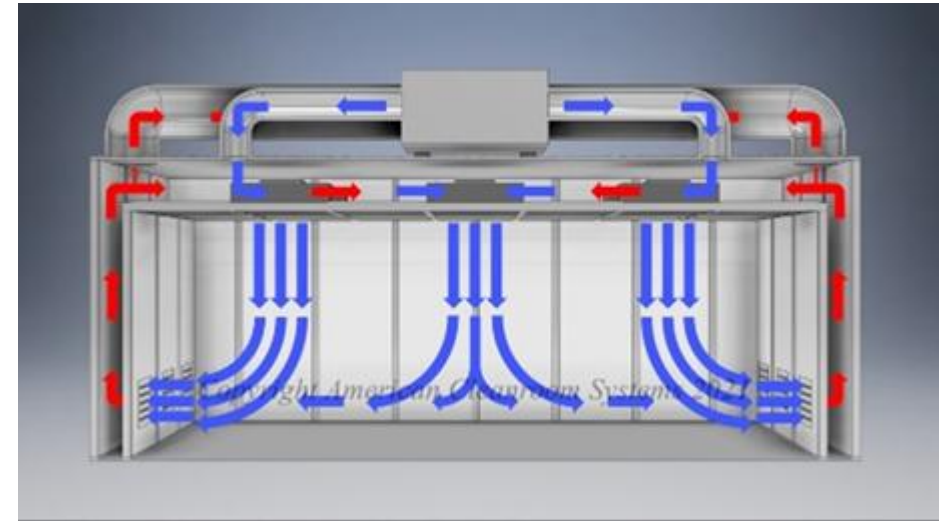
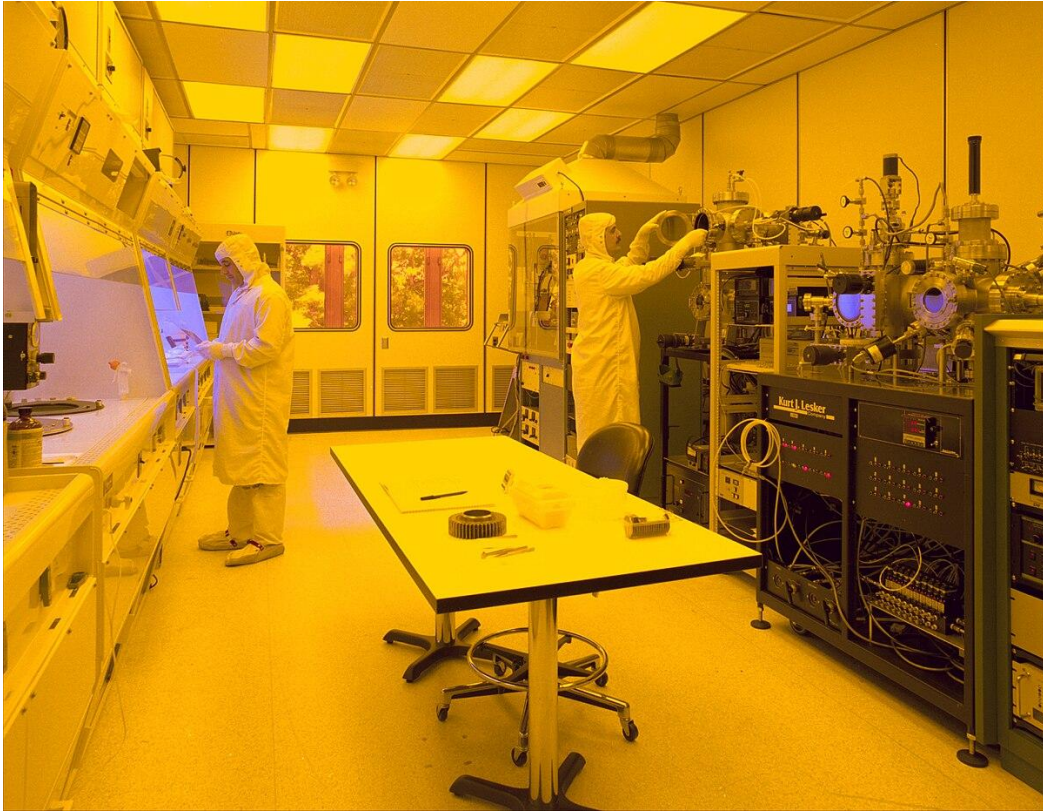


Industrial AC:-Water Curtains in Paint shops



Water Wash Paint Booth

Industrial AC:- Clean rooms



[Cleanroom - Wikipedia](https://en.wikipedia.org/wiki/Cleanroom)

According to Season

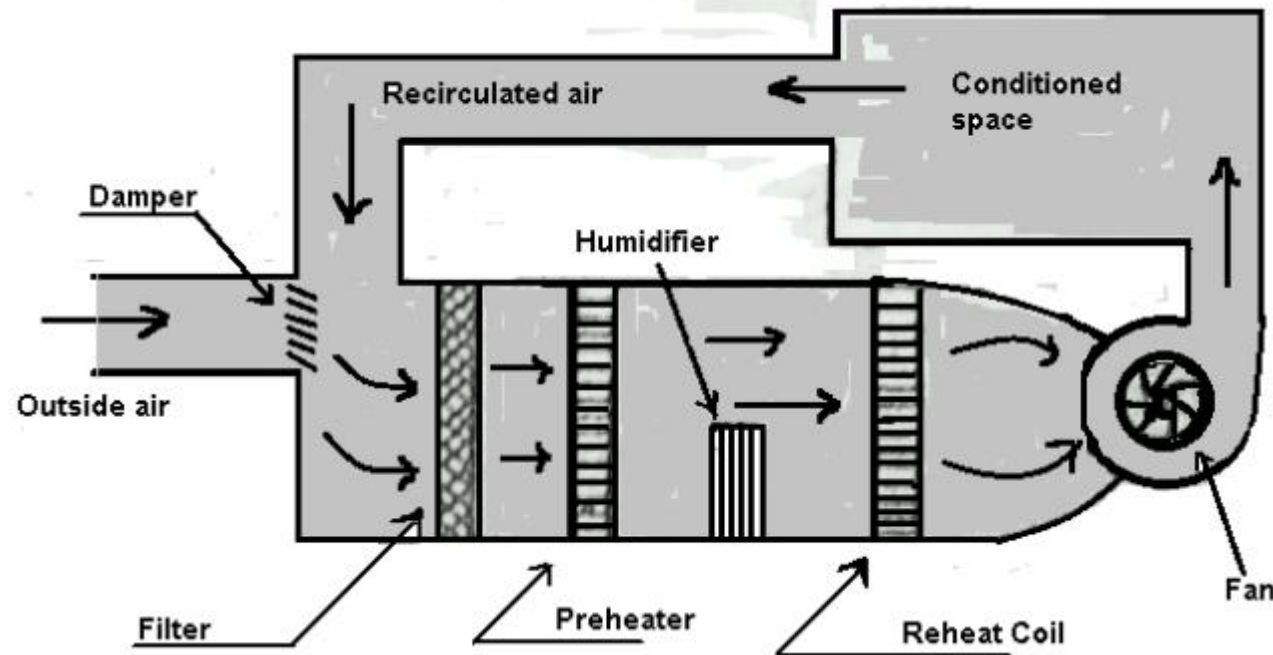


- Winter:- Heated and humidified
- Summer:- Cooled and dehumidified
- Year-round :- Both conditions depending on season

Winter Air-Conditioning

- Air is **heated and generally humidified**
- Outside air flows through the damper, and mixes up with recirculated air (obtained from the conditioned space).
- The mixed air passes through a filter to remove dust, dirt and other impurities
- Air now passes through a preheat coil in order to prevent the possible freezing of water and to control the evaporation of water in the humidifier
- The preheated air passes through a reheat coil to bring the air to the designed room temperature

Winter Air-Conditioning



Heating and Humidification

Winter Air-Conditioning

- The humidification of air can be achieved in several ways, e.g. by bringing the air in contact with a wetted surface, or with droplets of water as in an air washer, by adding water droplets directly to air or by direct addition of dry saturated or superheated steam
- The most common method of humidifying air for these applications is by direct addition of **dry steam to air**

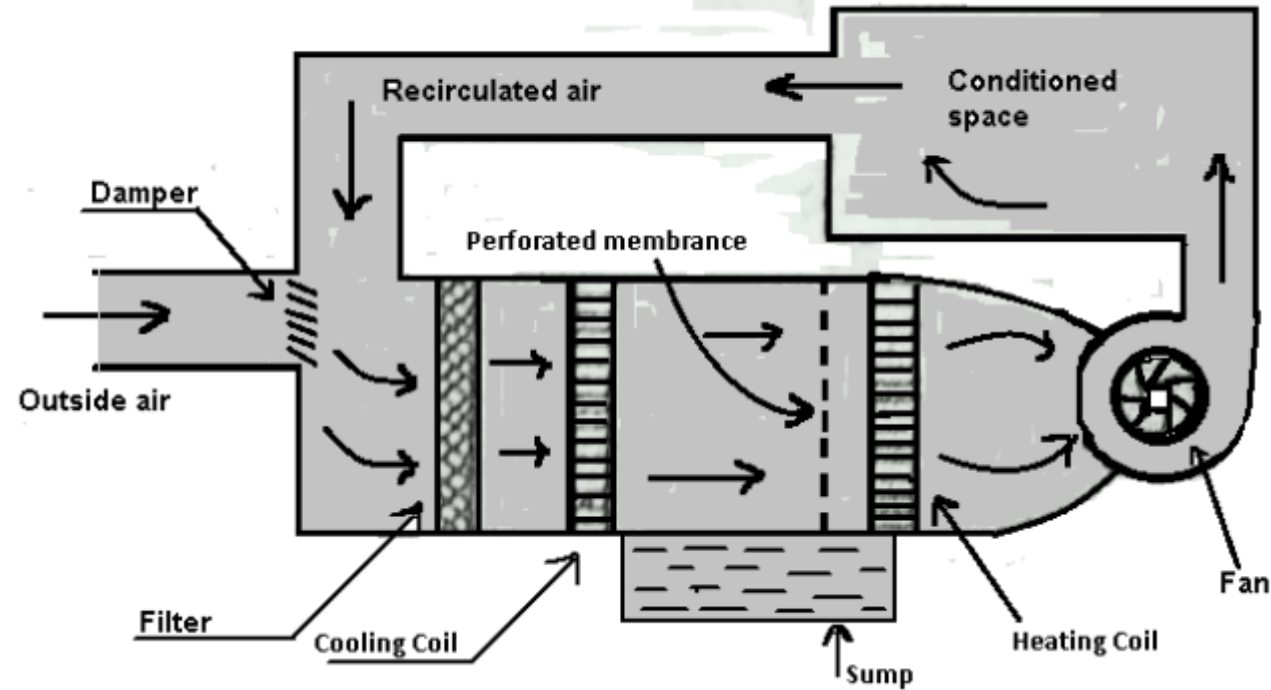


Dampers or valves to control outdoor indoor air flow

Summer Air-Conditioning

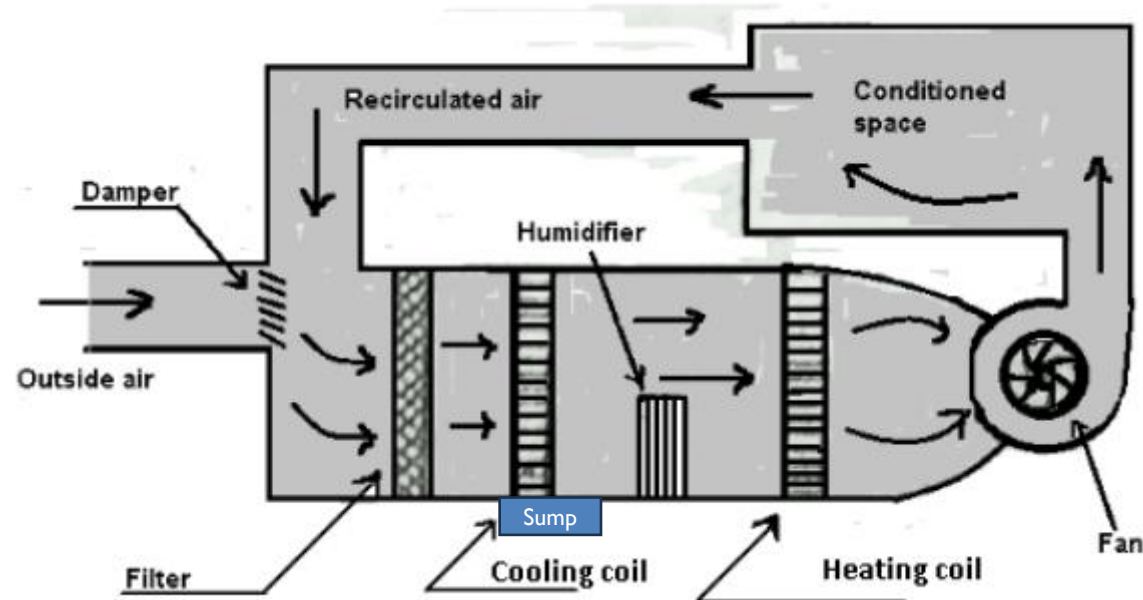
- Summer air conditioning system requires a refrigeration system that reduces the temperature of the air to temperatures much lower than the surroundings.
- Air is **cooled and generally dehumidified**
- Outside air flows through the damper, and mixes up with recirculated air (obtained from the conditioned space).
- The mixed air passes through a filter to remove dust, dirt and other impurities
- Air now passes through a cooling coil. The coil has a temperature much below the required dry bulb temperature of the air in the conditioned space
- The cooled air passes through a perforated membrane and loses its moisture in the condensed form which is collected in a sump
- Then air is heated by using a heating coil to bring the air to the designed room temperature

Summer Air-Conditioning



Cooling and Dehumidification

Year-Round Air Conditioning System



Summer + Winter A/C systems

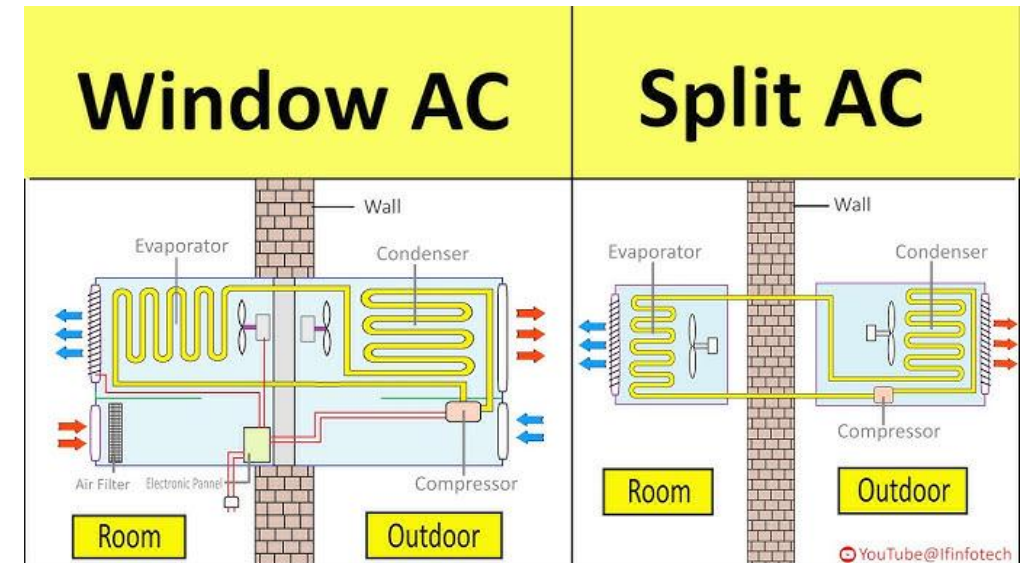
Summer:- cooling and dehumidification only active

Winter:- Heating and humidification only active

According To The Arrangement of Equipment

1. Unitary A/C

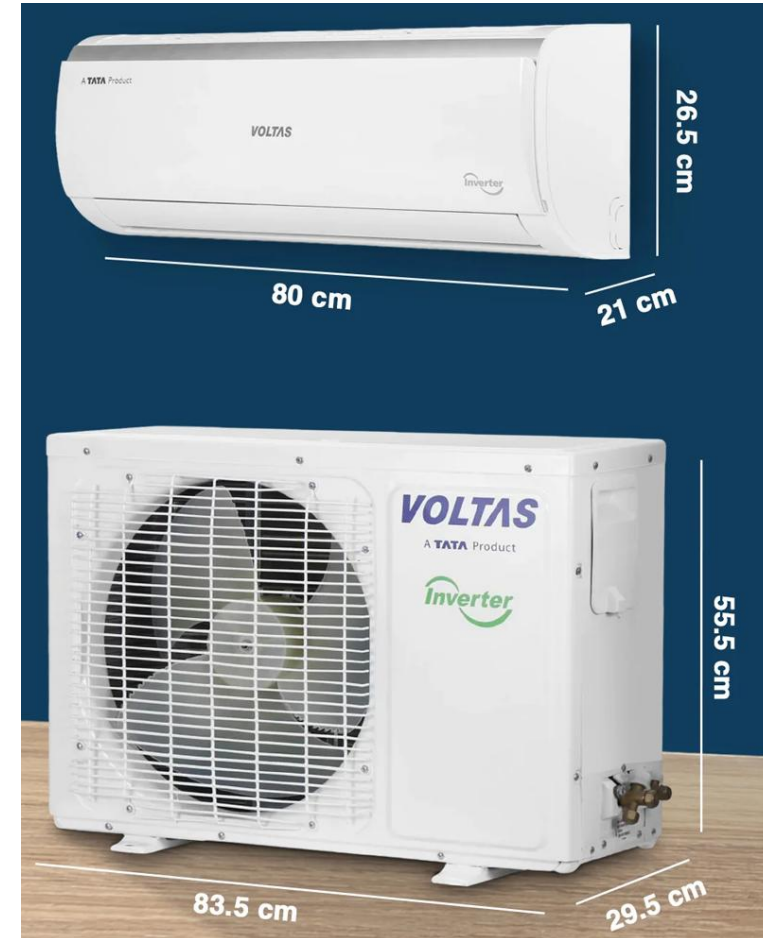
- Assembled A/C is installed in or adjacent to space to be conditioned
- Generally sit in a window or wall opening
- Interior air is cooled as a fan blows it over the evaporator
- Exterior air is heated as second fan flows it over the condenser
- Types
 - Window
 - Split (ductless system)



According To The Arrangement of Equipment

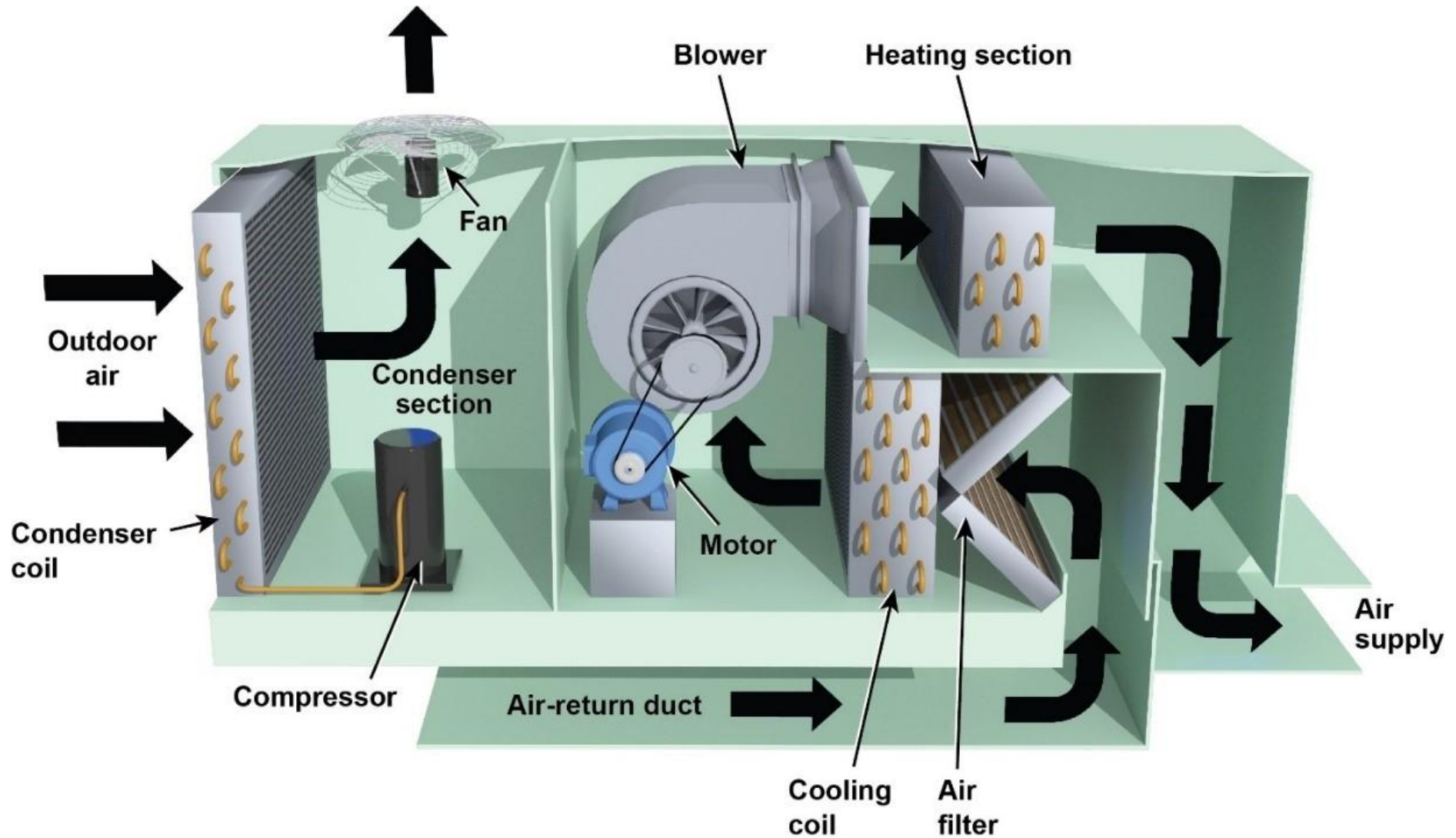


Window AC



Split AC

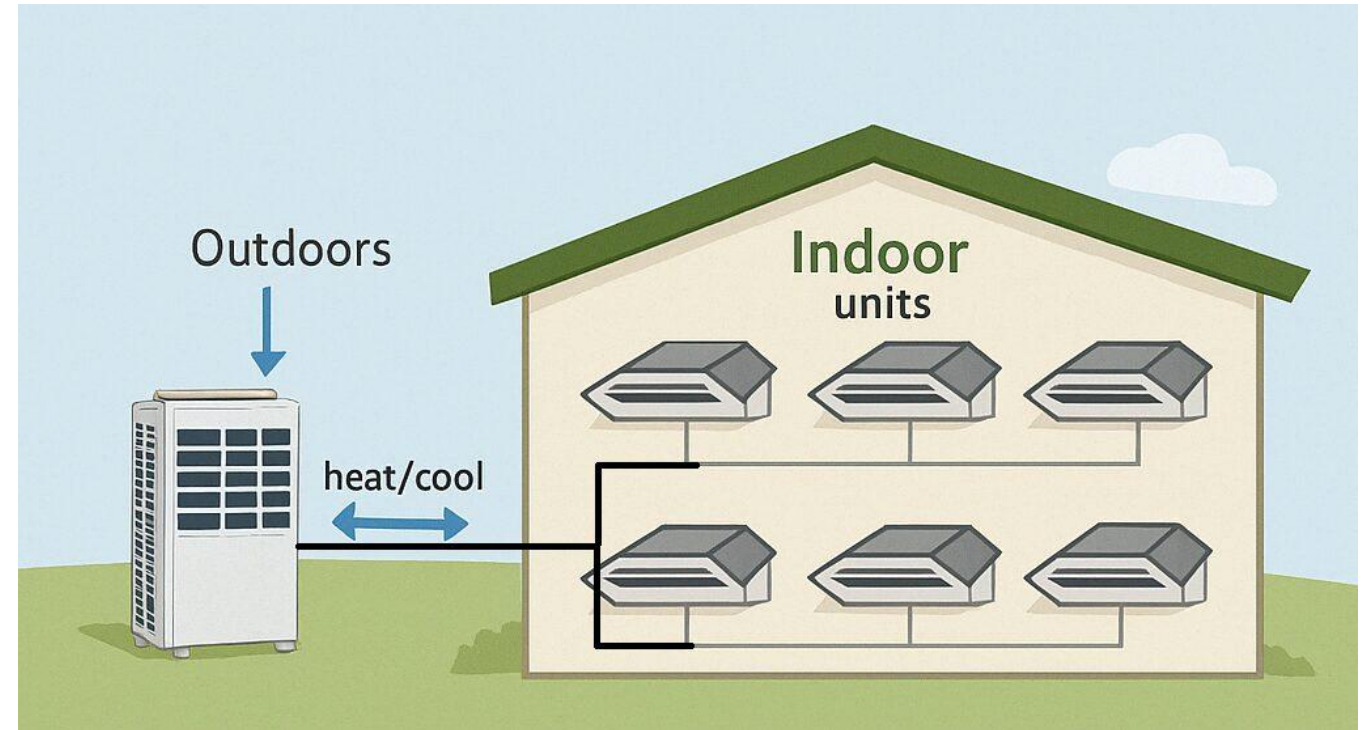
Unitary A/C



According To The Arrangement of Equipment

Central A/C

- Serve one or more thermal zones
- Major equipment is located outside the served zone
- Will have several control points such as thermostat for each zone
- Medium used can be air or water or both



Central Air-Conditioning System

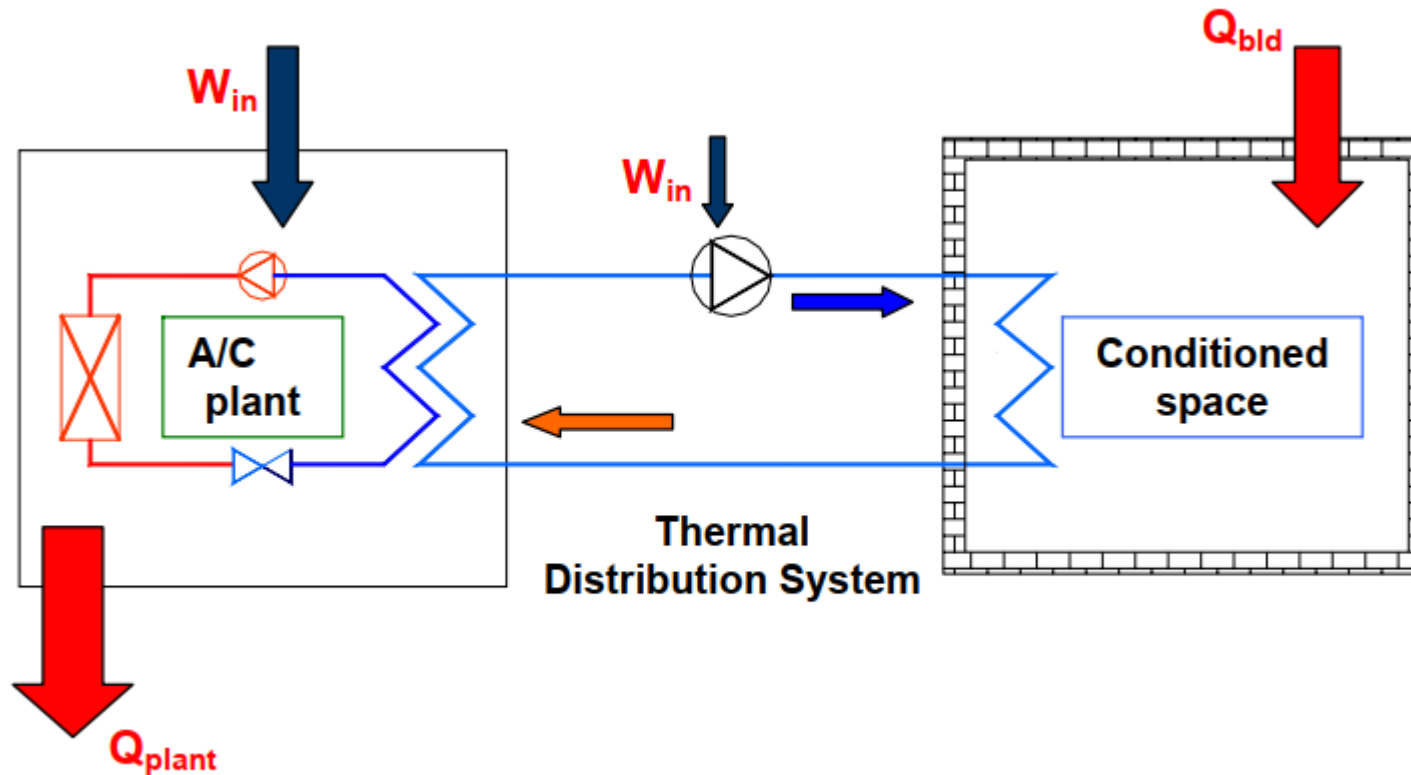
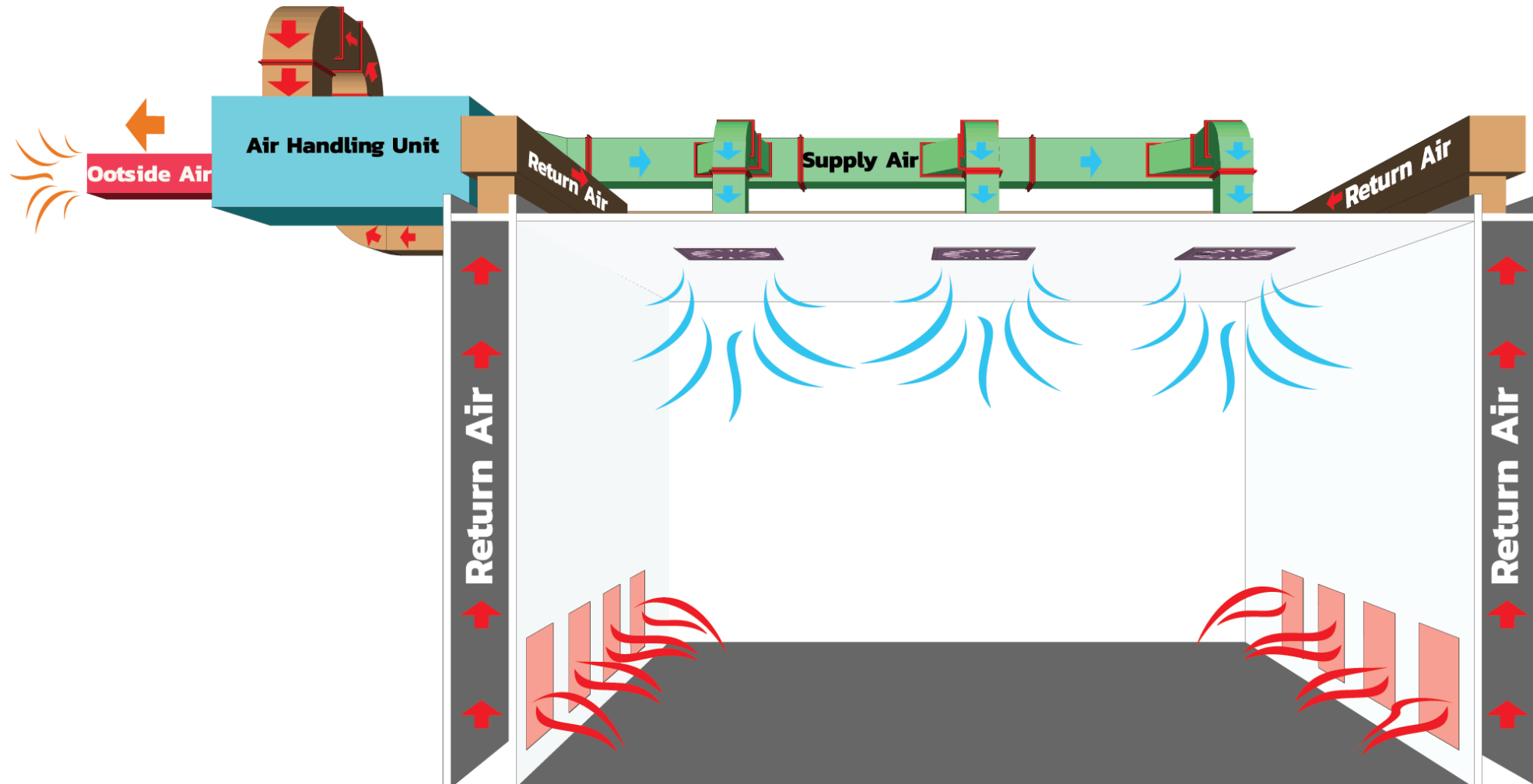
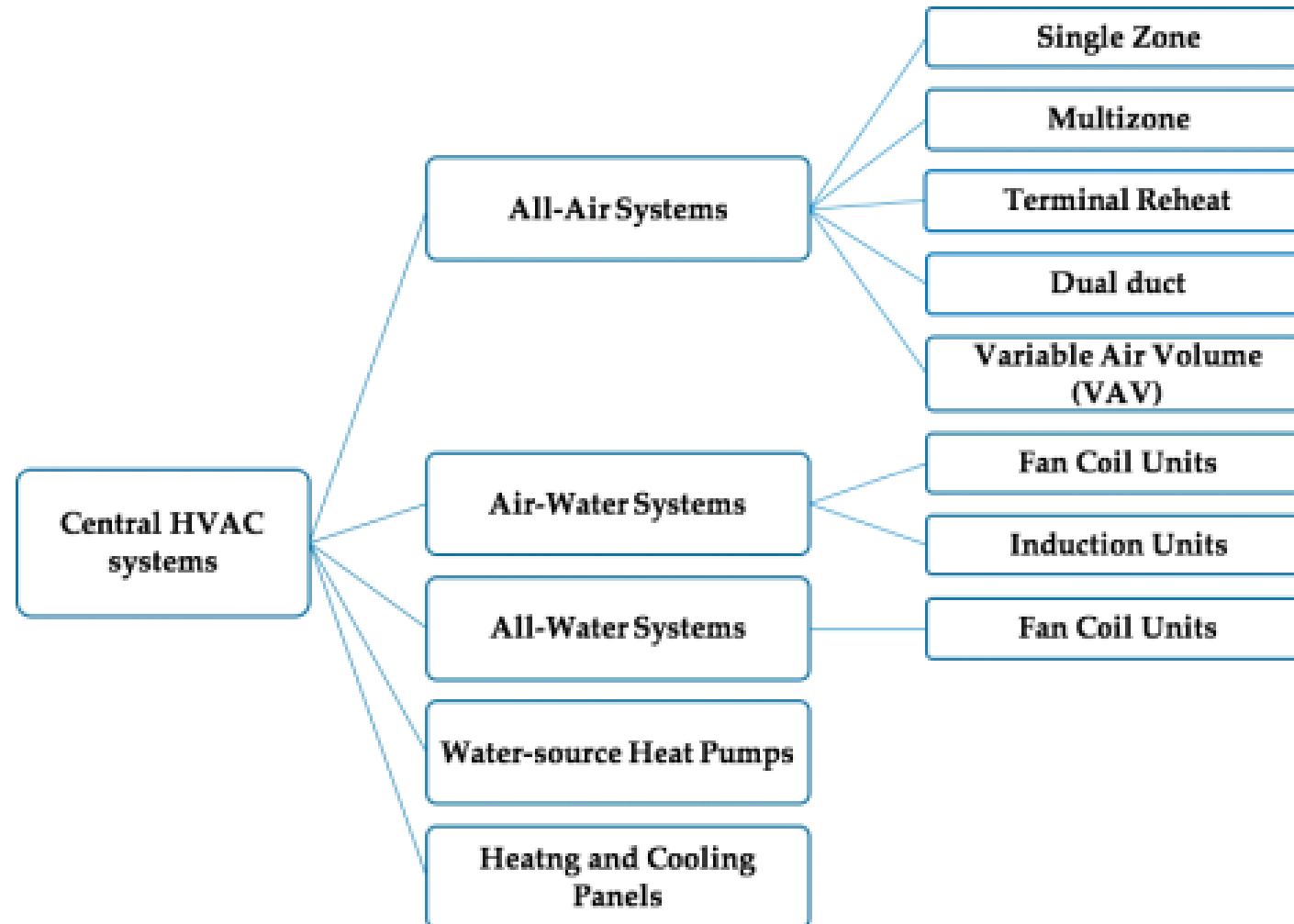


Fig.36.1: Schematic of a summer air conditioning system with the thermal distribution system

Central A/C system



Central AC



Classification of A/C systems



- All air system
- All water system
- Air-water system
- Unitary system

Air handling unit and zone system

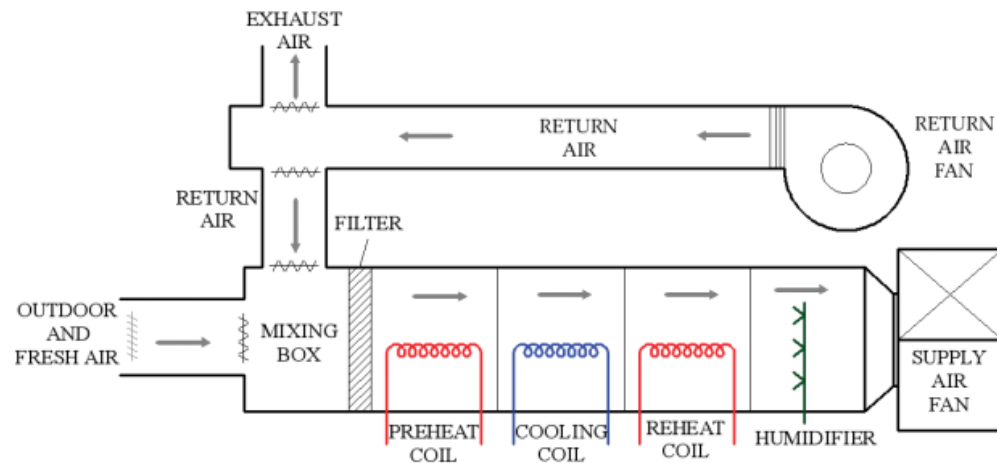


Figure 3. Equipment arrangement for central HVAC system.

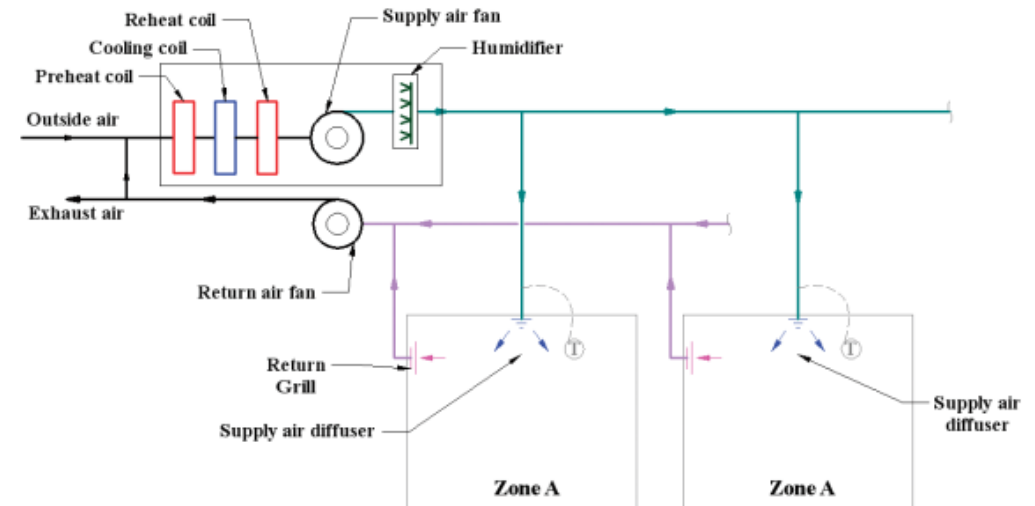


Figure 4. All-air HVAC system for single zone.

All Air system

- **Air is used as the media** that transports energy from the conditioned space to the A/C plant
- Air is then conveyed to the conditioned **space through insulated ducts using blowers and fans**
- Types
 - **Single duct:-** cooling and heating using same duct, but not both. Applications: Auditoriums, theatres etc.
 - **Dual duct:-** Supply air split into two stream viz. cooling and heating. Before reaching conditioning space this streams are mixed together to make req. temperature

All air system

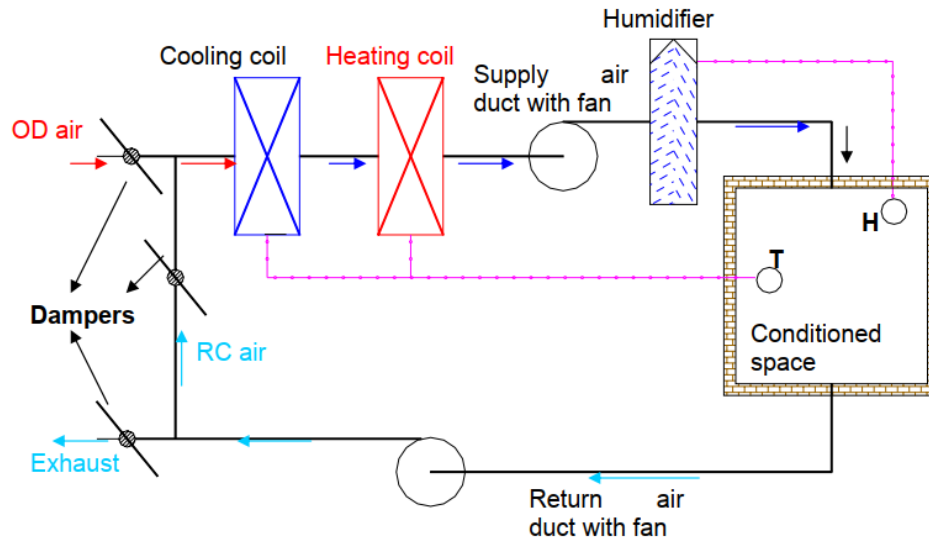


Fig.36.2. A constant volume, single zone system

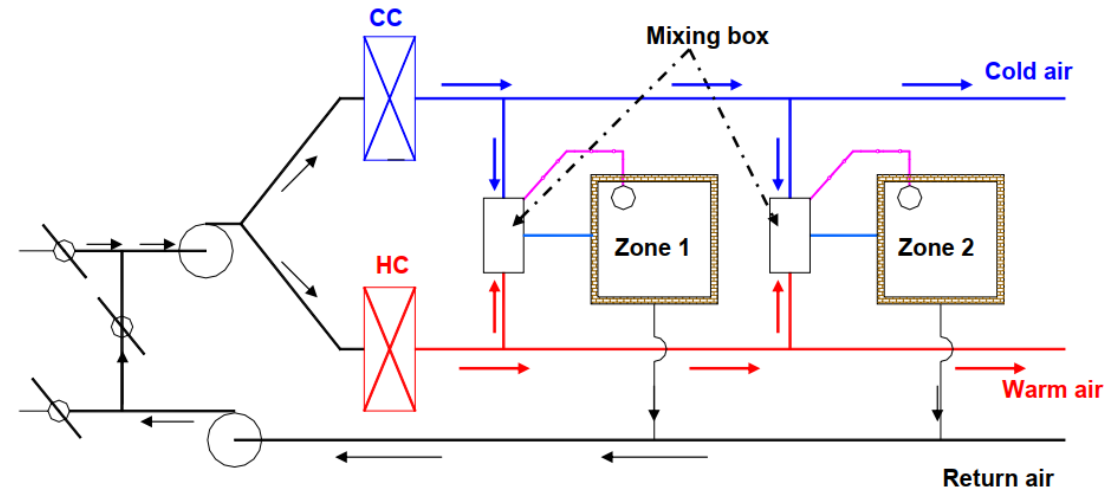


Fig.36.5: Dual duct, constant volume system

All Water system



- **Water transports energy between the conditioned space and the air conditioning plant**
- Provision must be there for supplying required amount of treated, outdoor air to the conditioned space for ventilation purposes
- Types
 - 2-Pipe system:- one for supply/one for return
 - 4-Pipe system:- one for cold water and one for hot water; and two return water pipelines
- Heat transfer between the cold/hot water and the conditioned space takes place either by convection, conduction or radiation or a combination of these.
- Applications: buildings requiring individual room control, such as hotels, apartment buildings and office buildings

All Water system

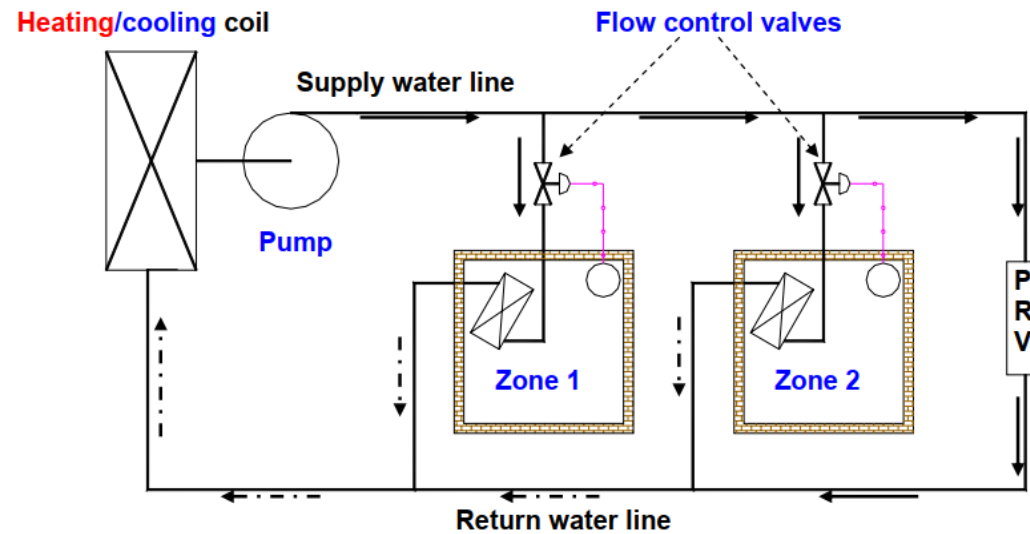


Fig.36.6: A two-pipe, all water system

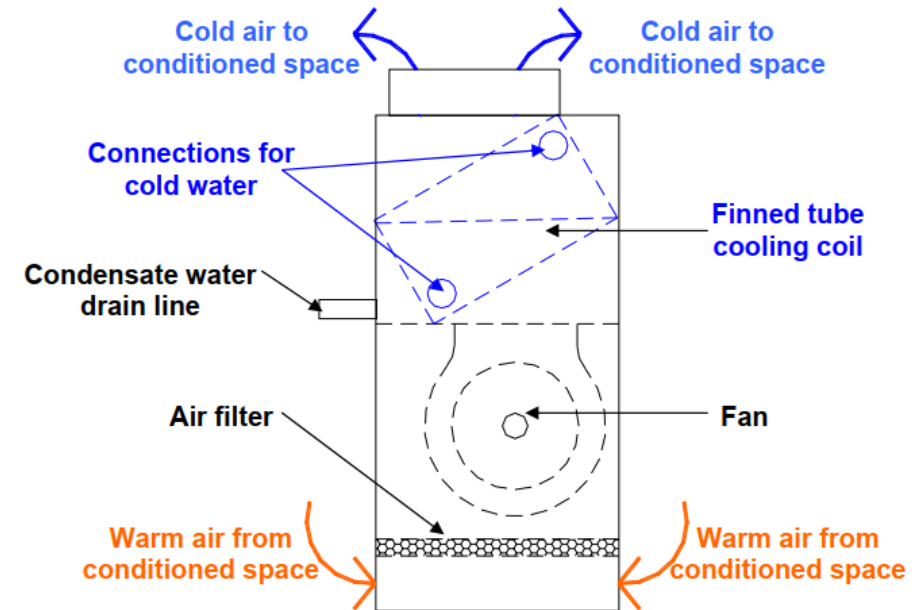


Fig.36.7: A basic fan coil unit for cooling purposes

Air-Water System



- Both air and water are used for providing required conditions in the conditioned space
- The air and water are cooled or heated in a central plant
- Primary Air:- air supplied to the conditioned space from the central plant
- Secondary Water:- water supplied from the plant
- Applications: exterior buildings with large sensible loads

Air-Water System

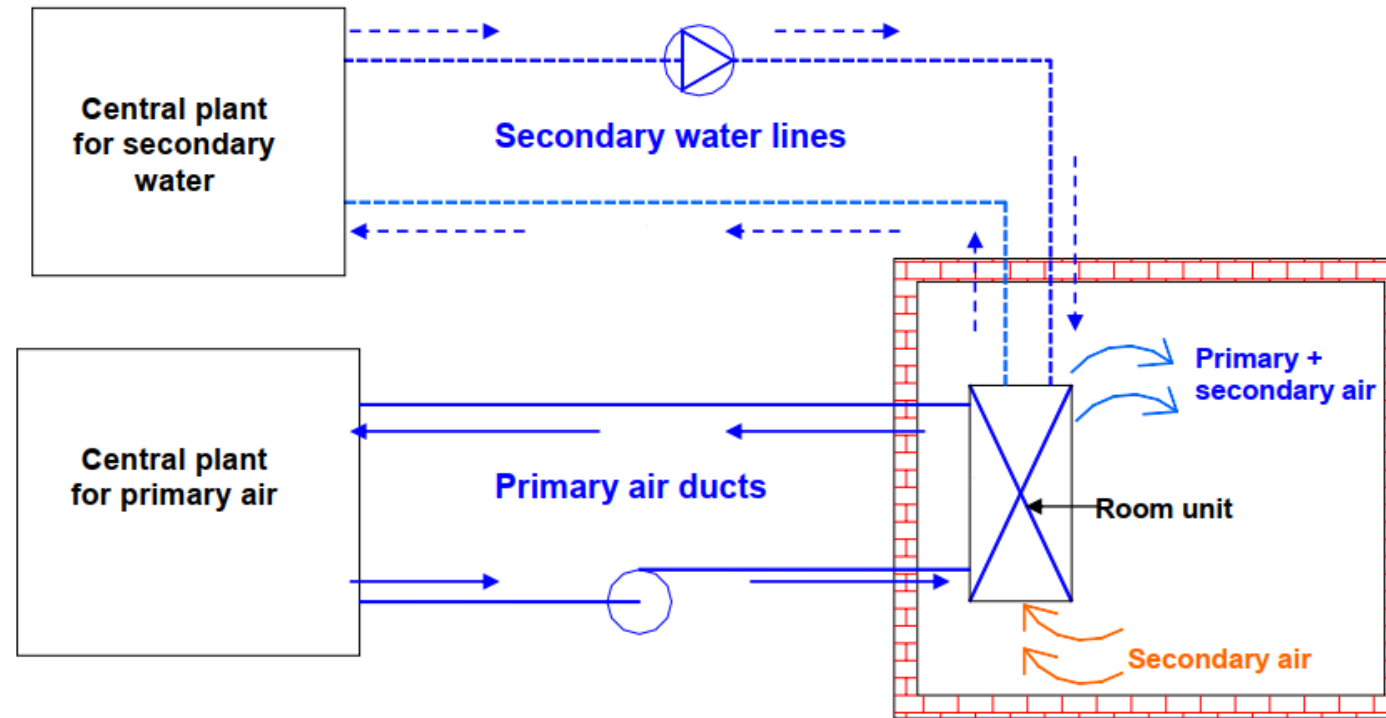


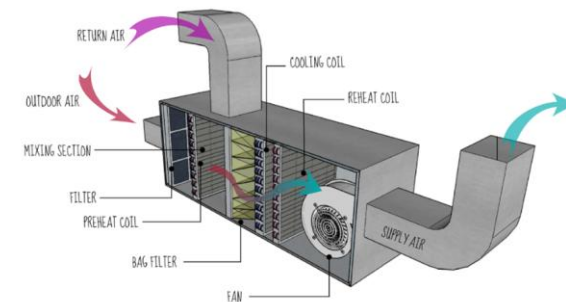
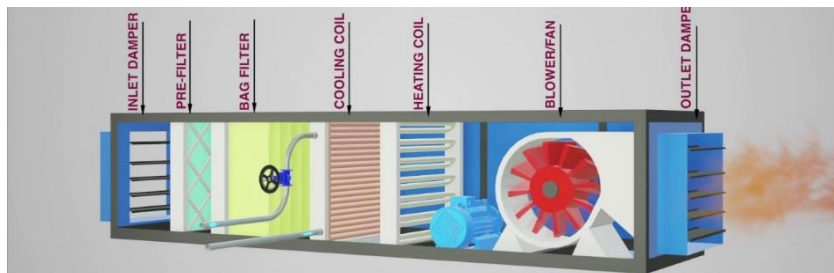
Fig.36.8: A basic air-water system

Air-Flow Through Ducts

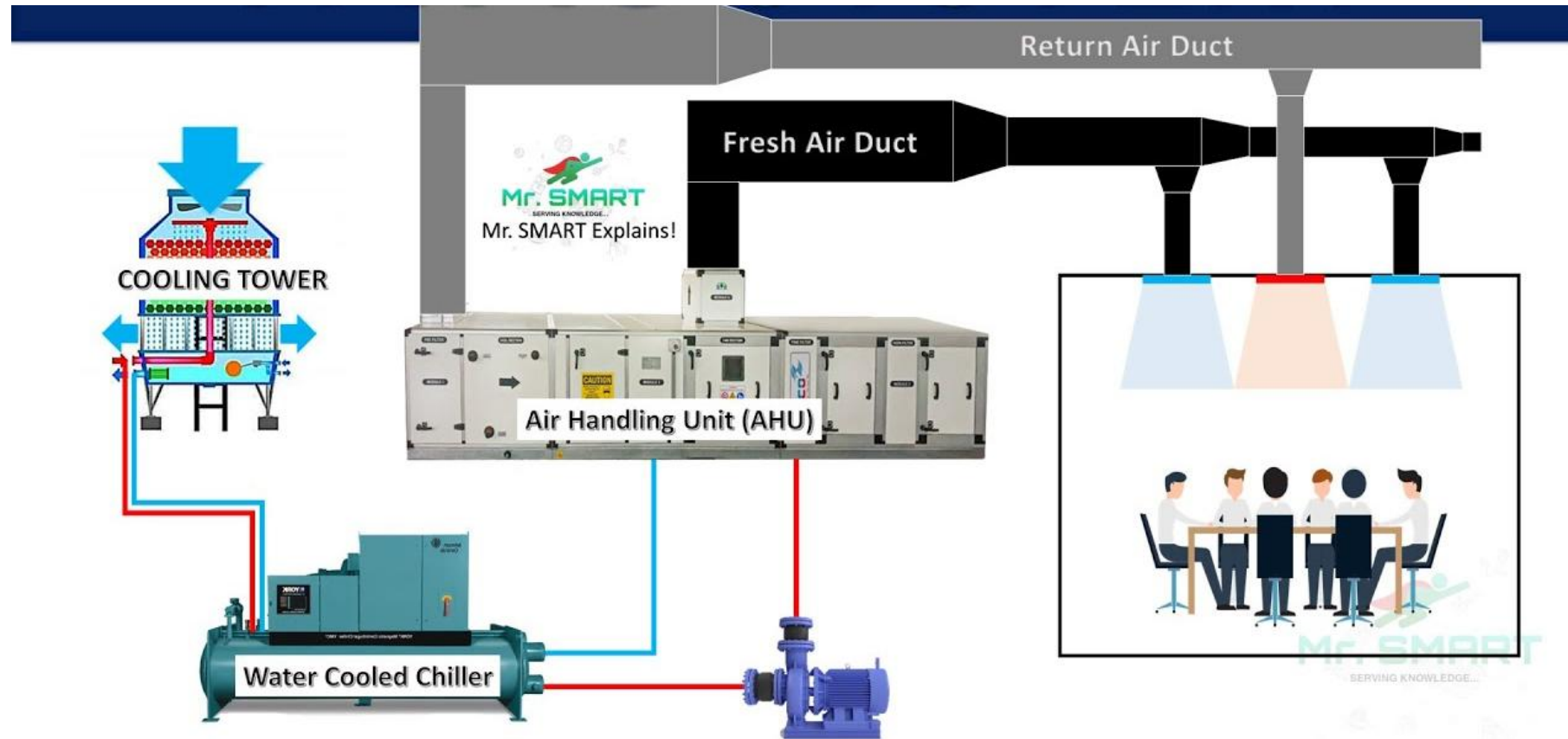


Air-Handling Unit (AHU)

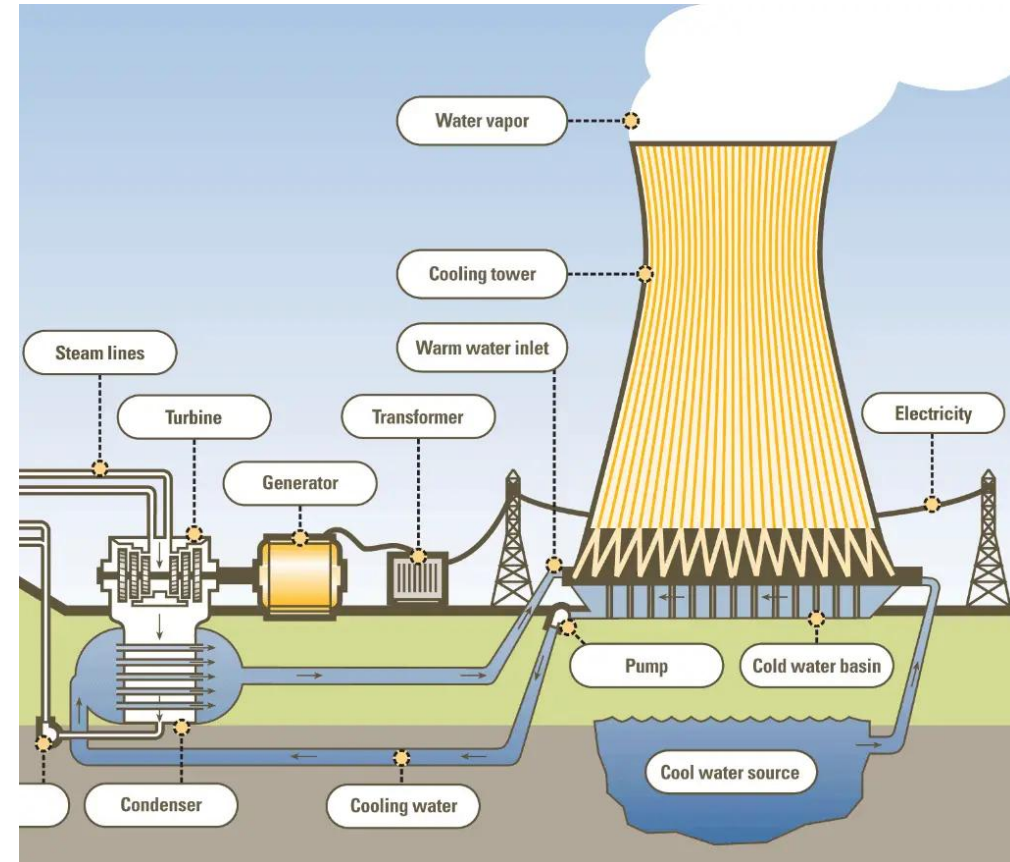
- To transmit processed air from the air conditioning plant to the conditioned space and distribute it properly within the conditioned space
- It consist of;
 1. A duct system that includes a **supply air duct, return air duct, cooling and/or heating coils, humidifiers /dehumidifiers, air filters and dampers**
 2. An air distribution system comprising **various types of outlets for supply air and inlets for return air**
 3. Supply and return air fans which provide the necessary energy to move the air throughout the system



Air-Flow Through Ducts

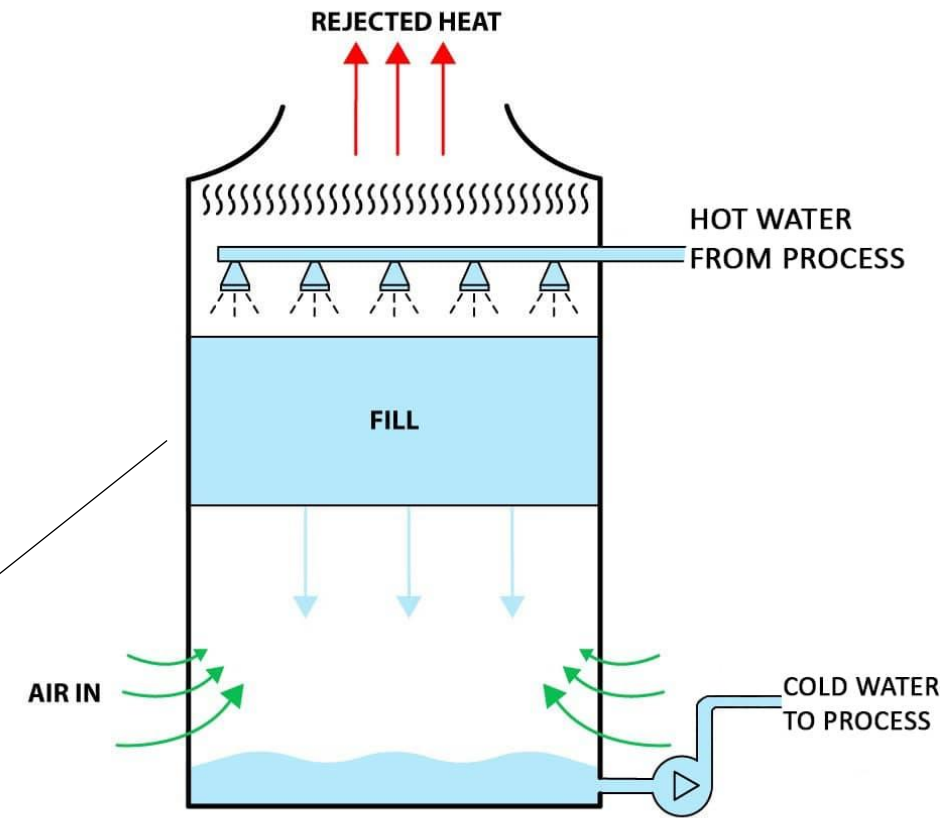


Cooling Tower:- Natural



These types of cooling towers are generally used in Power Plants

Cooling Tower:- Induced



These types of cooling towers are generally used in ACs

Flow through Ducts

- Fan motor consumes a large amount of power, and the duct system occupies considerable building space, the design of air transmission system is an important step in the complete design of air conditioning systems
- Flow of air through ducts
 - Major loss:
 - Minor loss:

Pressure Loss in Ducts

- Mainly due to
 - Friction losses
 - Momentum change due to change of direction and/or velocity

The pressure drop due to friction is known as **frictional pressure drop or friction loss, Δp_f** . The pressure drop due to momentum change is known as **momentum pressure drop or dynamic loss, Δp_d** . Thus the total pressure drop Δp_t is given by:

$$\Delta p_t = \Delta p_f + \Delta p_d \quad (37.6)$$

- Frictional losses: Darcy-Weisbach equation or empirical correlation suggested by Colebrook and White

$$\Delta p_f = f \frac{L}{D} \left(\frac{\rho V^2}{2} \right)$$

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left[\frac{k_s}{3.7D} + \frac{2.51}{(Re_D)\sqrt{f}} \right]$$

Dynamic losses

- Dynamic pressure loss takes place whenever there is a change in either the velocity or direction of airflow due to the use of a variety of bends and fittings in air
- In general these fittings and bends are rather short in length (< 1 m), the major pressure drop as air flows through these fittings is not because of viscous drag (friction) but due to momentum change

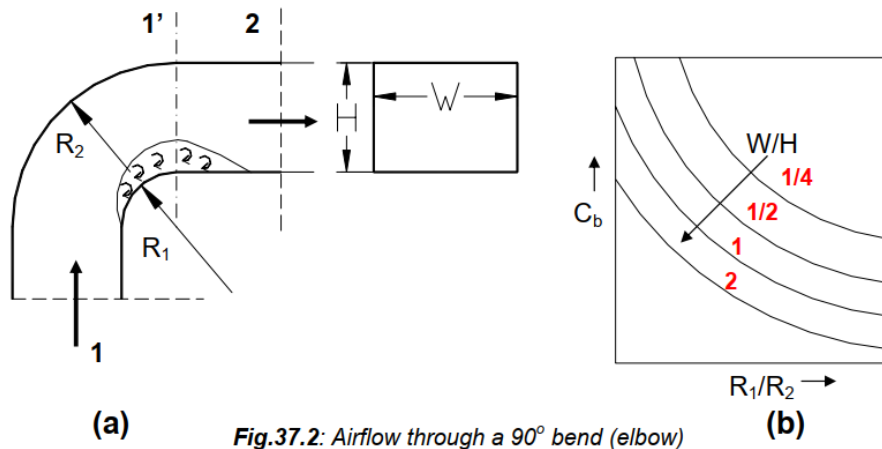


Fig.37.2: Airflow through a 90° bend (elbow)

$$\Delta p_{d,b} = C_b \left(\frac{\rho V^2}{2} \right) = f((W/H), R_1, R_2) \left(\frac{\rho V^2}{2} \right)$$

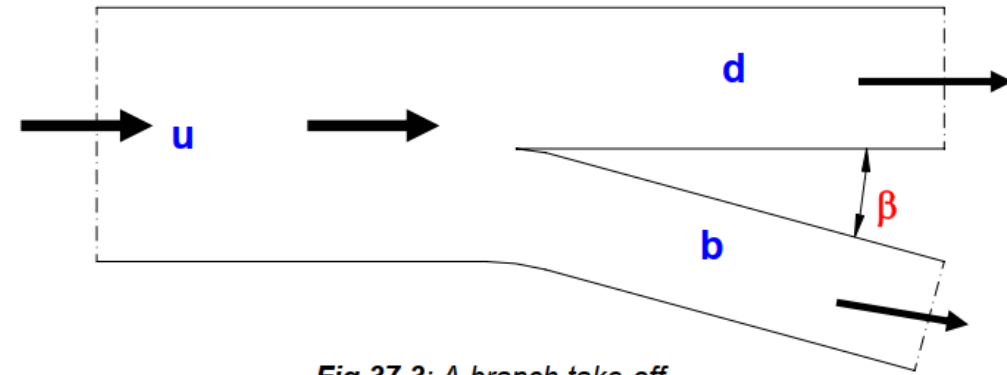


Fig.37.3: A branch take-off

$$\Delta p_{u-d} = 0.4 \left(\frac{\rho V_d^2}{2} \right) \left(1 - \frac{V_d}{V_u} \right)^2 \quad (37.15)$$

where V_d and V_u are the air velocities in the downstream and upstream ducts, respectively.

The dynamic pressure drop from the upstream (u) to branch (b), Δp_{u-b} is given by:

$$\Delta p_{u-b} = C_{u-b} \left(\frac{\rho V_d^2}{2} \right) \quad (37.16)$$

Dynamic losses...

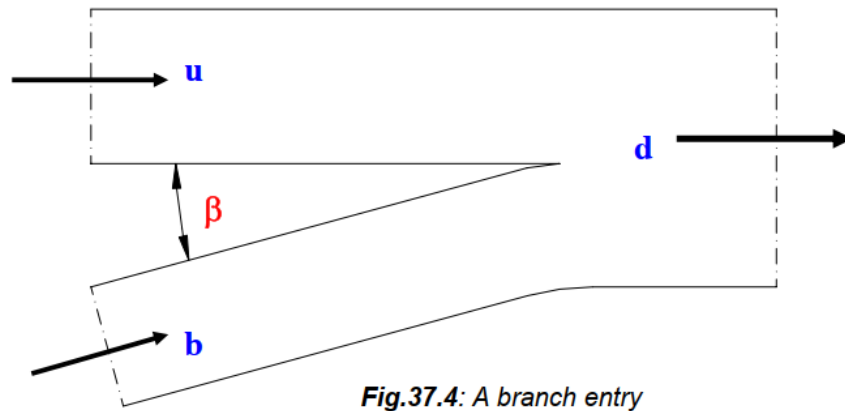


Fig.37.4: A branch entry

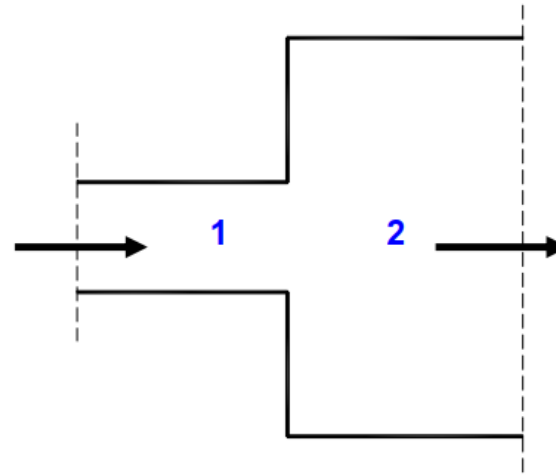


Fig.37.5(a): Sudden enlargement

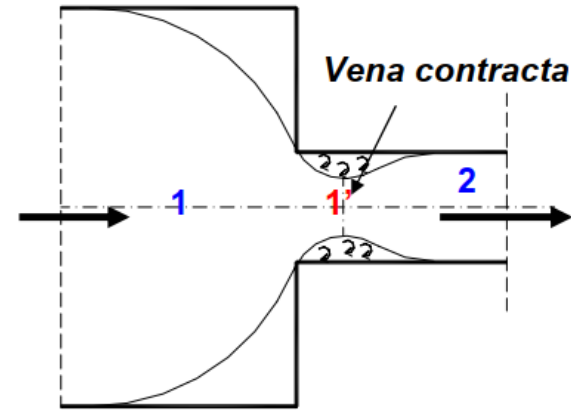


Fig.37.5(b): Sudden contraction

Using tables

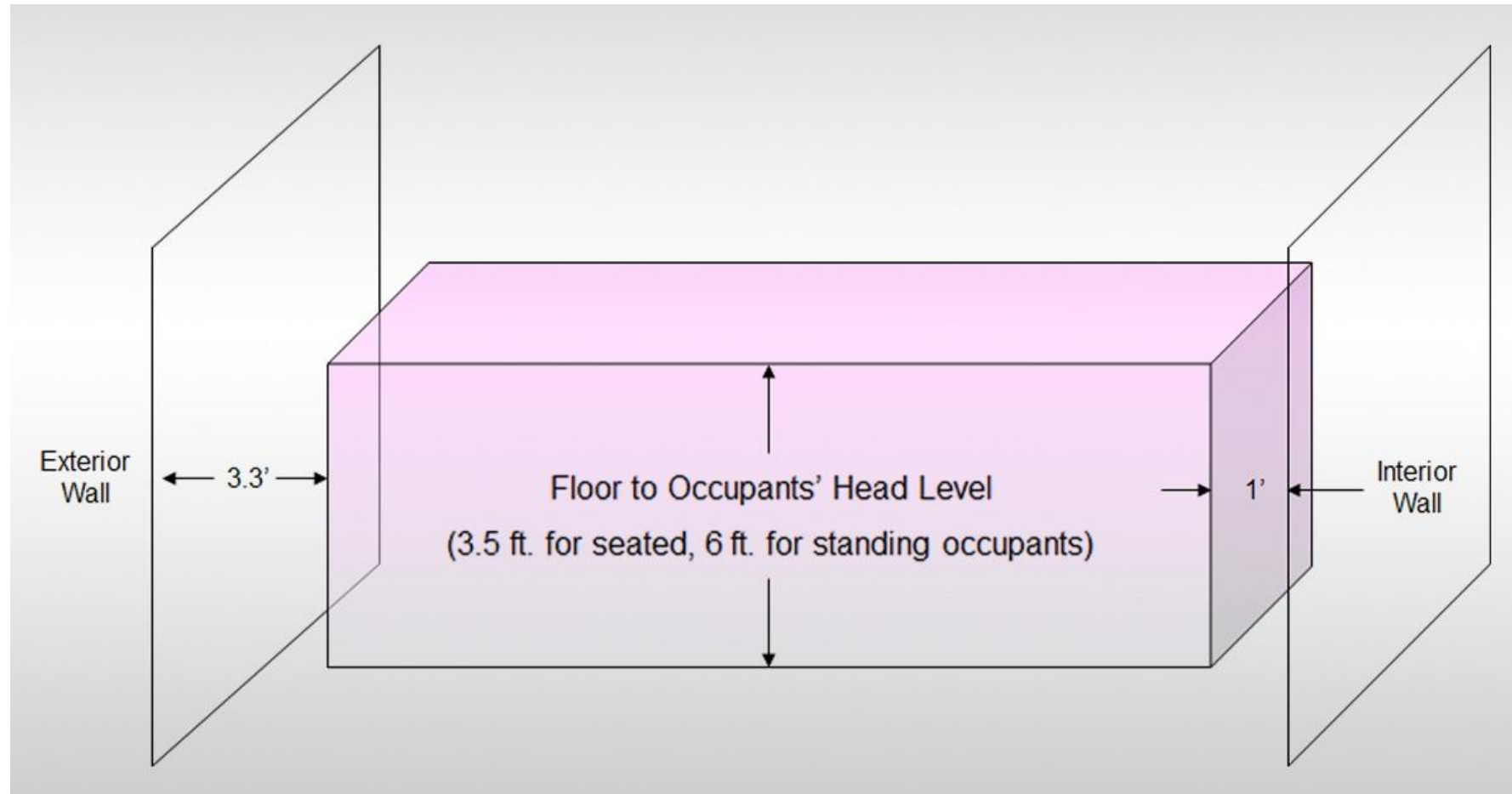
$$\Delta p_{d, \text{enl}} = \left(\frac{\rho V_1^2}{2} \right) \left(1 - \frac{A_1}{A_2} \right)^2$$

$$\Delta p_{d, \text{con}} = \left(\frac{\rho V_2^2}{2} \right) \left(\frac{A_2}{A_1} - 1 \right)^2 = \left(\frac{\rho V_2^2}{2} \right) \left(\frac{1}{C_c} - 1 \right)^2$$

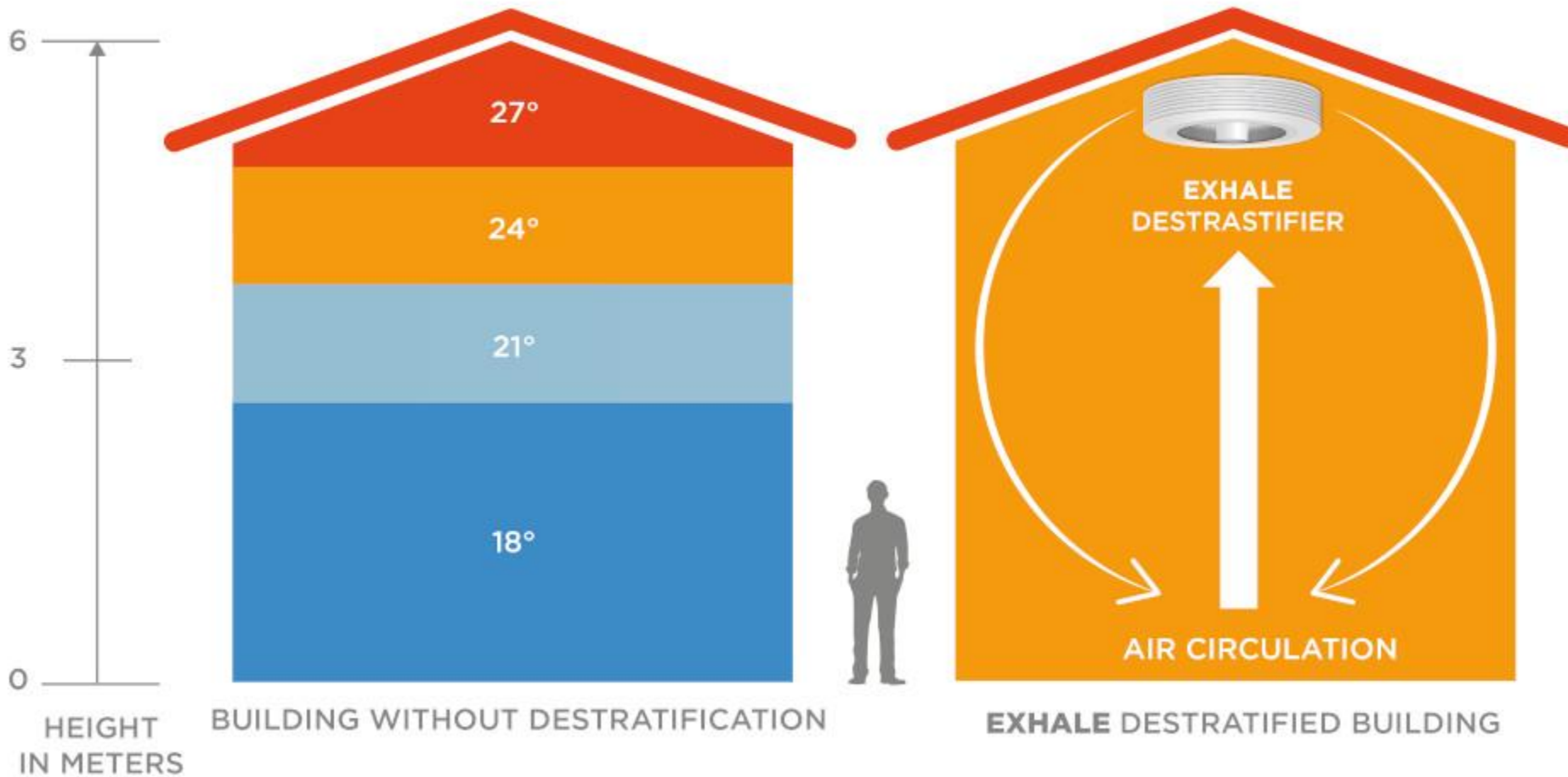
A_2/A_1	C_c
0.1	0.624
0.5	0.681
0.8	0.813
1.0	1.000

Thank You

Occupied Zone



Air Stratification



[Air destratification - Exhale Fans](#)

Space air distribution



Factors

- Supply air velocity and temperature
- Geometry and position of air supply outlet
- Position of return air inlet
- Room Geometry
- Internal heat source eg: people, appliances
- Room turbulence

Space air distribution..

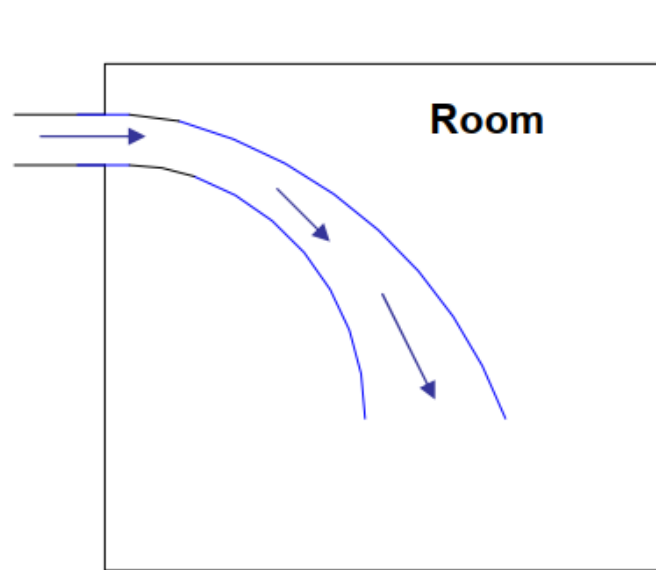


Fig.39.1(a): Drop of a cool air jet

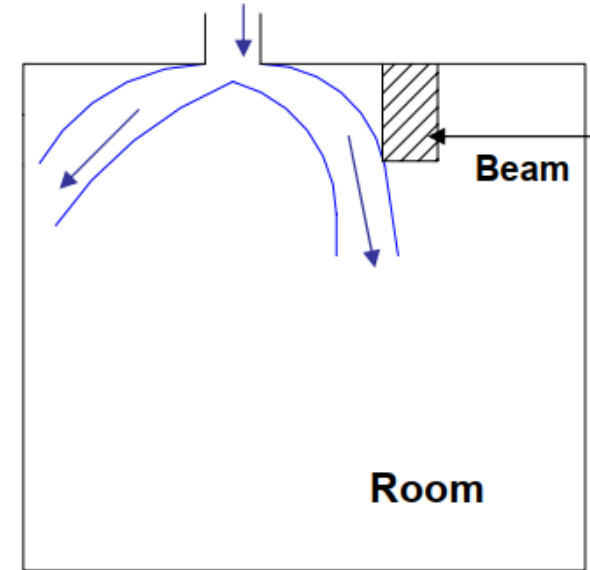


Fig.39.1(b): Deflection of a cool air jet

Space air distribution:- diffuser geometry

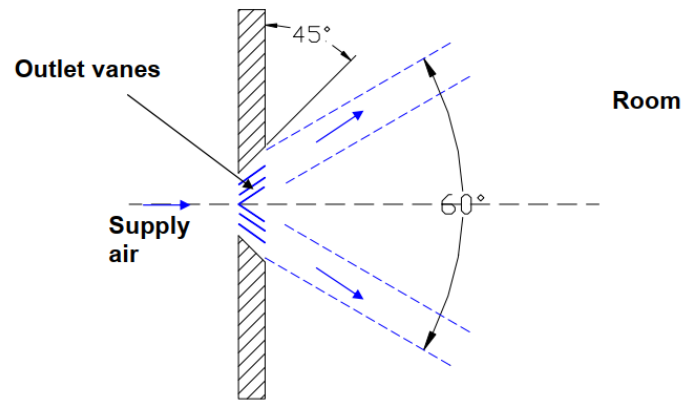


Fig.39.3: Spread of an air jet with diverging vanes

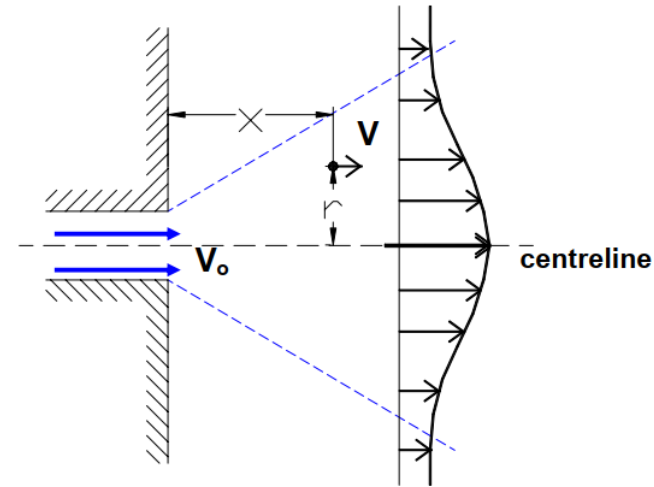


Fig.39.4: Velocity distribution through a circular jet

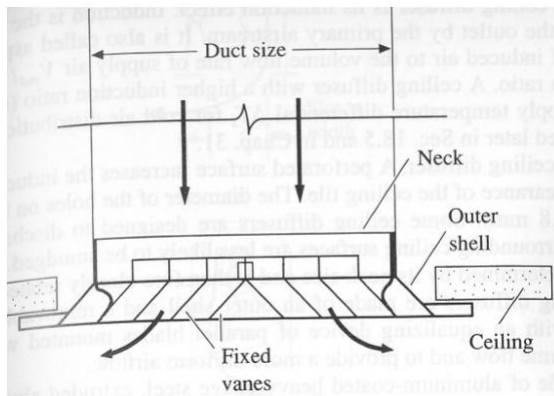
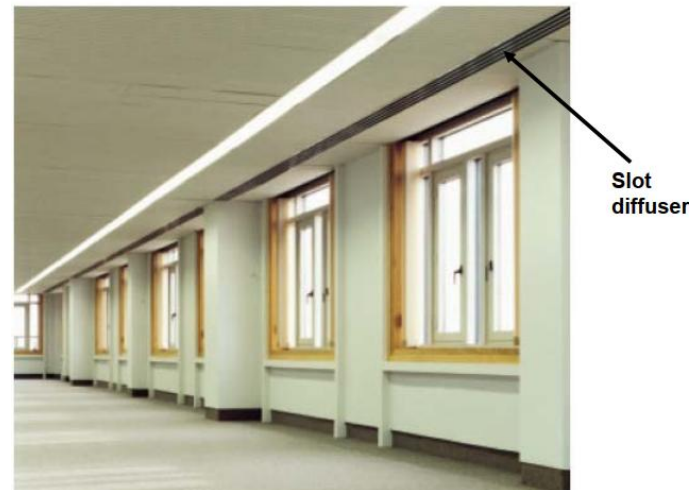


Fig.39.6(a): Schematic of a ceiling diffuser



Space air distribution..

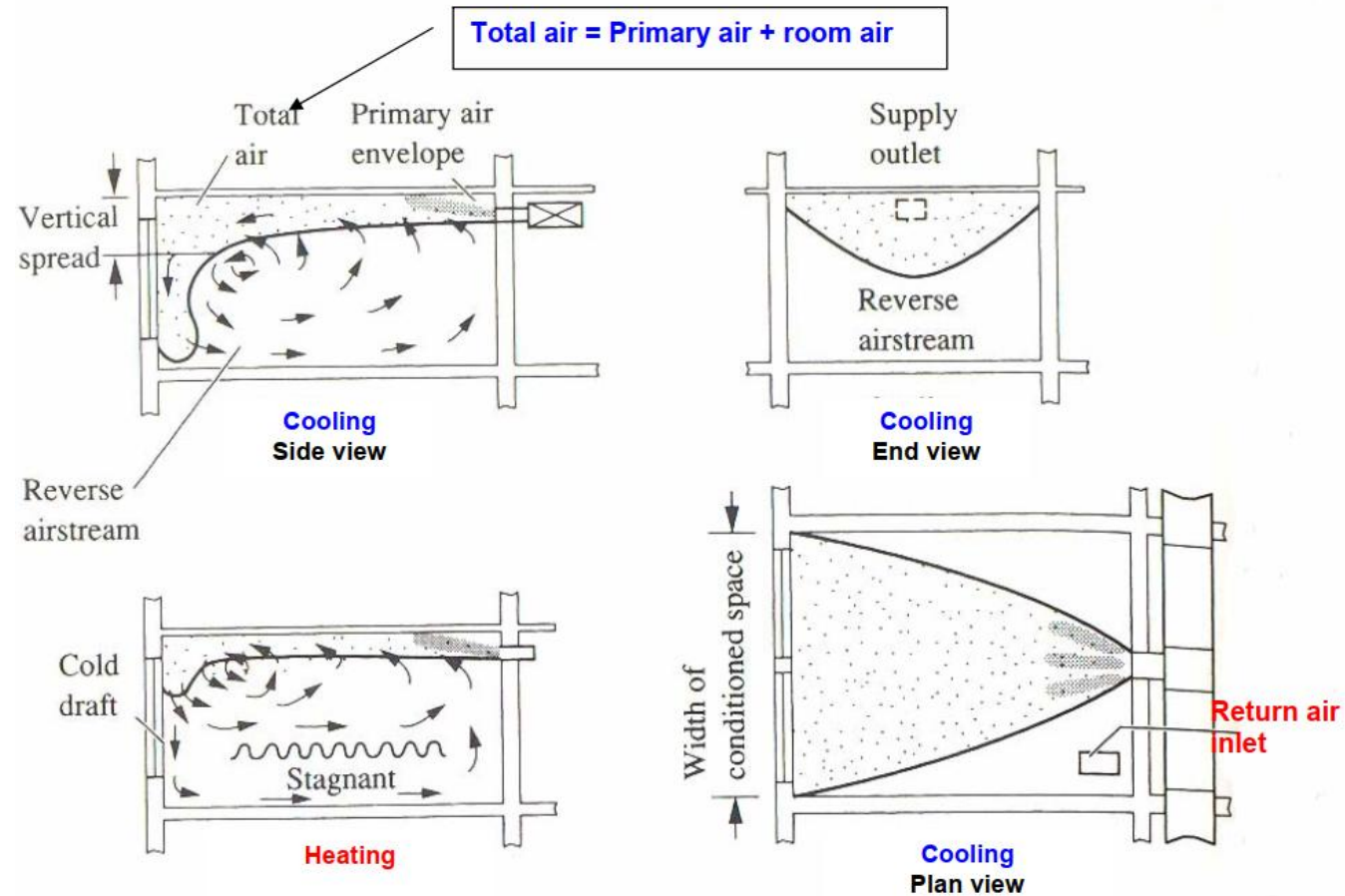


Fig.39.9: Airflow pattern using high side outlets for cooling and heating applications

Reference

<https://archive.nptel.ac.in/courses/112/105/112105129/>

